CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 21: Trigonometrical Identities

Previous year board questions, topic-wise: 13 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 The three basic identities

1. Assertion (A): If $\sin^2 A + \sin A = 1$, then $\cos^4 A + \cos^2 A = 1$. Reason (R): $1 - \sin^2 A = \cos^2 A$. (a) A true, R false (b) A false, R true (c) both true, and R is the correct reason for A (d) both true, but R is not the correct reason for A ICSE 2025, 1 mark

2. Statement 1: $\sin^2\theta + \cos^2\theta = 1$. Statement 2: $\operatorname{cosec}^2\theta + \cot^2\theta = 1$. Which is valid? (a) only 1 (b) only 2 (c) both (d) neither ICSE 2024, 1 mark

3. $(1 + \sin A)(1 - \sin A)$ equals: (a) $\operatorname{cosec}^2 A$ (b) $\sin^2 A$ (c) $\sec^2 A$ (d) $\cos^2 A$ ICSE 2023, 1 mark

2 Proving identities

4. Prove that $\dfrac{(\cot A + \tan A - 1)(\sin A + \cos A)}{\sin^3 A + \cos^3 A} = \sec A \operatorname{cosec} A$. ICSE 2025, 4 marks

5. Factorise $\sin^3\theta + \cos^3\theta$. Hence prove that $\dfrac{\sin^3\theta + \cos^3\theta}{\sin\theta + \cos\theta} + \sin\theta\cos\theta = 1$. ICSE 2024, 4 marks

6. Prove that $(\sin^2\theta - 1)(\tan^2\theta + 1) + 1 = 0$. ICSE 2023, 4 marks

7. Prove that $\dfrac{\sin A}{1 + \cot A} - \dfrac{\cos A}{1 + \tan A} = \sin A - \cos A$. ICSE 2020, 3 marks

8. Prove that $\left(\dfrac{1 - \tan\theta}{1 - \cot\theta}\right)^2 = \tan^2\theta$. ICSE 2020, 4 marks

9. Prove that $(\operatorname{cosec}\theta - \sin\theta)(\sec\theta - \cos\theta)(\tan\theta + \cot\theta) = 1$. ICSE 2019, 3 marks

10. Prove that $\sqrt{\sec^2\theta + \operatorname{cosec}^2\theta} = \tan\theta + \cot\theta$. ICSE 2018, 3 marks

11. Prove that $(1 + \cot\theta - \operatorname{cosec}\theta)(1 + \tan\theta + \sec\theta) = 2$. ICSE 2018, 4 marks

12. Prove that $\dfrac{\sin\theta - 2\sin^3\theta}{2\cos^3\theta - \cos\theta} = \tan\theta$. ICSE 2017, 3 marks

3 Complementary angles

13. Without using tables, evaluate $\sin^2 28^\circ + \sin^2 62^\circ + \tan^2 38^\circ - \cot^2 52^\circ + \dfrac{1}{4}\sec^2 30^\circ$. ICSE 2017, 3 marks

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.

MathSetu